

## Call by value and **call by reference**/address:

### Passing parameters to the functions: [parameter passing techniques]

In C-Language, we can send the arguments to the functions in 2 ways.

1. **Call by value / pass by value.**
2. **Call by address / pass by address. [ call by reference ]**

### Call by value / pass by value:

In call by value we are sending actual parameter value to the formal parameter. Later there is no relation is maintained in between actual and formal parameters. Due to this any change in formal parameter doesn't effects the value of actual parameter.

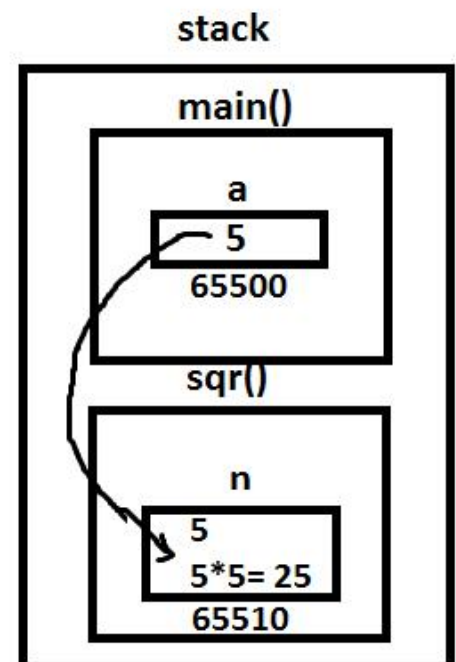
Eg: 1

```
#include<stdio.h>
#include<conio.h>

void sqr(int n)
{
    n = n * n;
} /* n deleted after the function execution */

void main()
{
```

### CALL BY VALUE



```

int a=5;
clrscr();
printf("Before function call a = %d\n" ,a);
sqr(a); /* fun calling */
printf("After function call a = %d", a);
getch();
}

```

### **Output:**

Before function call a = 5

After function call a = 5

### **Eg: 2 swapping of two integers**

```

#include<stdio.h>
#include<conio.h>
void swap(int a, int b)
{
int temp=a;
a=b;
b=temp;
}

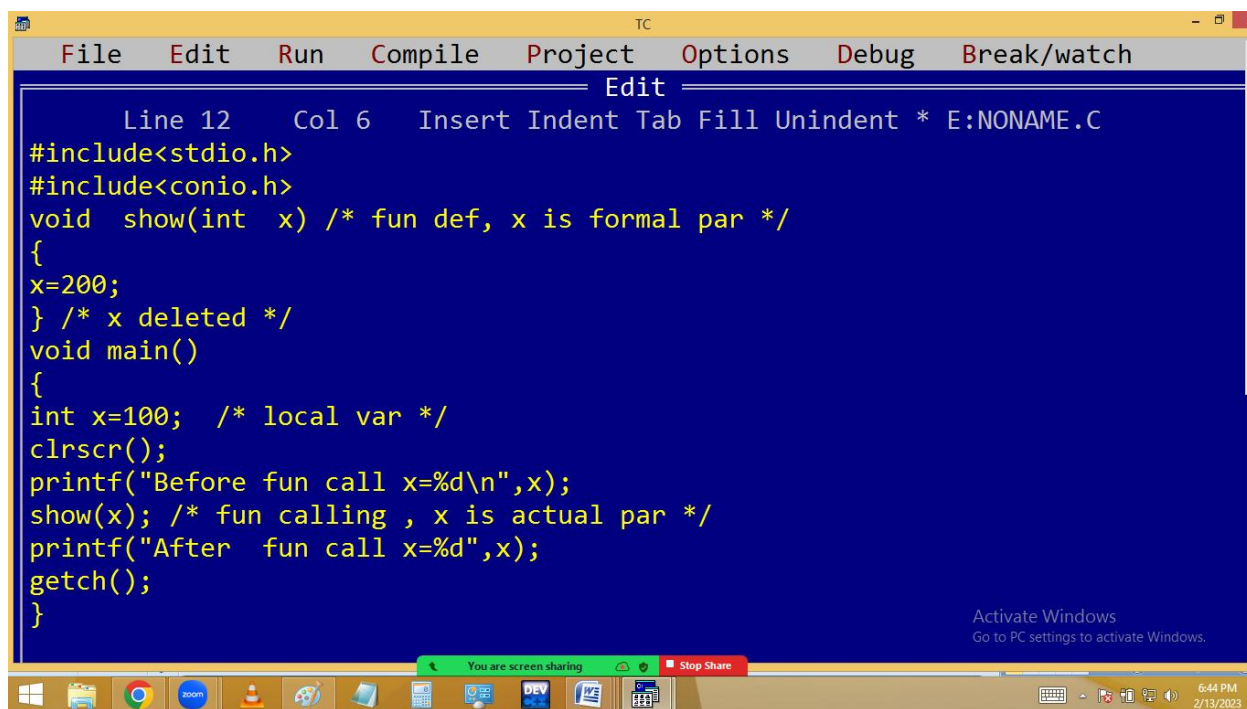
```

```
void main()
{
int a=5, b=7;
clrscr();
printf("Before fun call a=%d, b=%d\n" , a , b);
swap(a, b);
printf("After fun call a=%d, b=%d", a , b);
getch();
}
```

**Output:**

Before fun call a=5, b=7

After fun call a=5, b=7



```
TC
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Edit
Line 12 Col 6 Insert Indent Tab Fill Unindent * E:NONAME.C
#include<stdio.h>
#include<conio.h>
void show(int x) /* fun def, x is formal par */
{
x=200;
} /* x deleted */
void main()
{
int x=100; /* local var */
clrscr();
printf("Before fun call x=%d\n",x);
show(x); /* fun calling , x is actual par */
printf("After fun call x=%d",x);
getch();
}

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```

```
Before fun call x=100
After fun call x=100
```

TC

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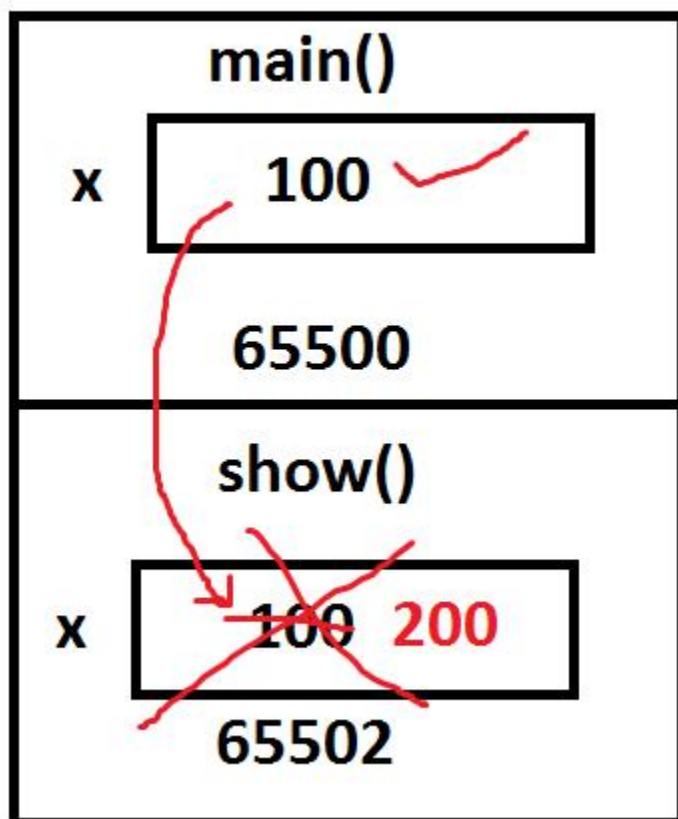
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## Call by /pass By value



## Call by address [Reference]:

In call by address, the address of actual parameter is passed to formal parameter. Due to this the formal parameter should be declared as a pointer. Then only the formal parameter receives the actual parameter address. Due to this any changes in formal parameter effects in actual parameter address i.e. actual parameter value.

Hence pointers allows the local variables to access outside the functions and this process is called call by address / reference.

It is very much useful in handling the strings, arrays etc outside the functions.

Eg: 1

```
#include<stdio.h>
#include<conio.h>
```

```
void sqr(int *n)
```

```
{
```

```
*n = *n * *n;
```

```
}
```

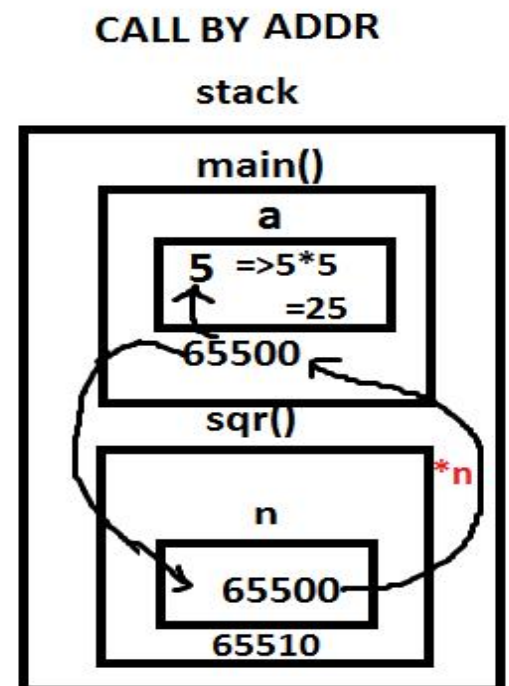
```
void main()
```

```
{
```

```
int a=5;
```

```
clrscr();
```

```
printf("Before function call a = %d\n " ,
```



```
a);  
sqr(&a); /* fun calling with address */  
printf("After function call a = %d " , a);  
getch();  
}
```

### **Output:**

Before function call a = 5

After function call a = 25

### **Eg: 2 Swap of two integers**

```
#include<stdio.h>  
  
#include<conio.h>  
  
void swap(int *a, int *b)  
{  
    int temp=*a; *a = *b; *b=temp;  
}  
  
void main()  
{  
    int a=5, b=7;  
    clrscr();
```

```
printf("Before fun call a=%d, b=%d\n", a ,b);
swap(&a, &b);
printf("After fun call a=%d, b=%d", a ,b);
getch();
}
```

### **Output:**

Before function call a=5, b=7

After function call a=7, b=5

```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 9 Col 28 Insert Indent Tab Fill Unindent * E:NONAME.C
#include<stdio.h>
#include<conio.h>
void show(int * x) /* fun def, x is formal par */
{
*x=200;
} /* x deleted */
void main()
{
int x=100; /* local var */_
clrscr();
printf("Before fun call x=%d\n",x);
show(&x); /* fun calling , x is actual par */
printf("After fun call x=%d",x);
getch();
}
```

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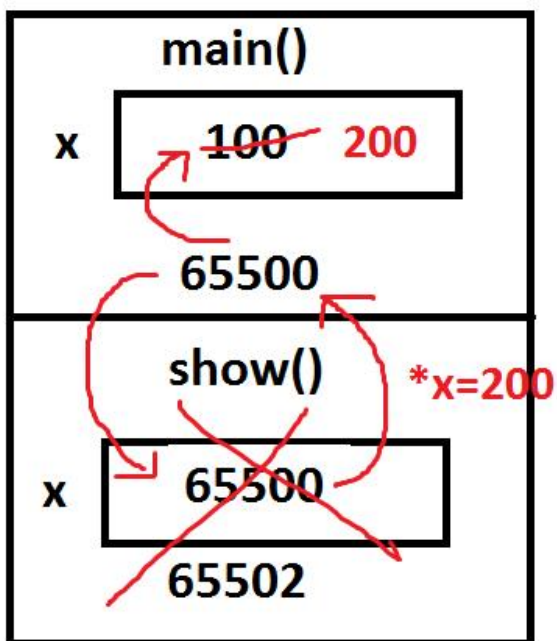
```
Before fun call x=100
After  fun call x=200
```

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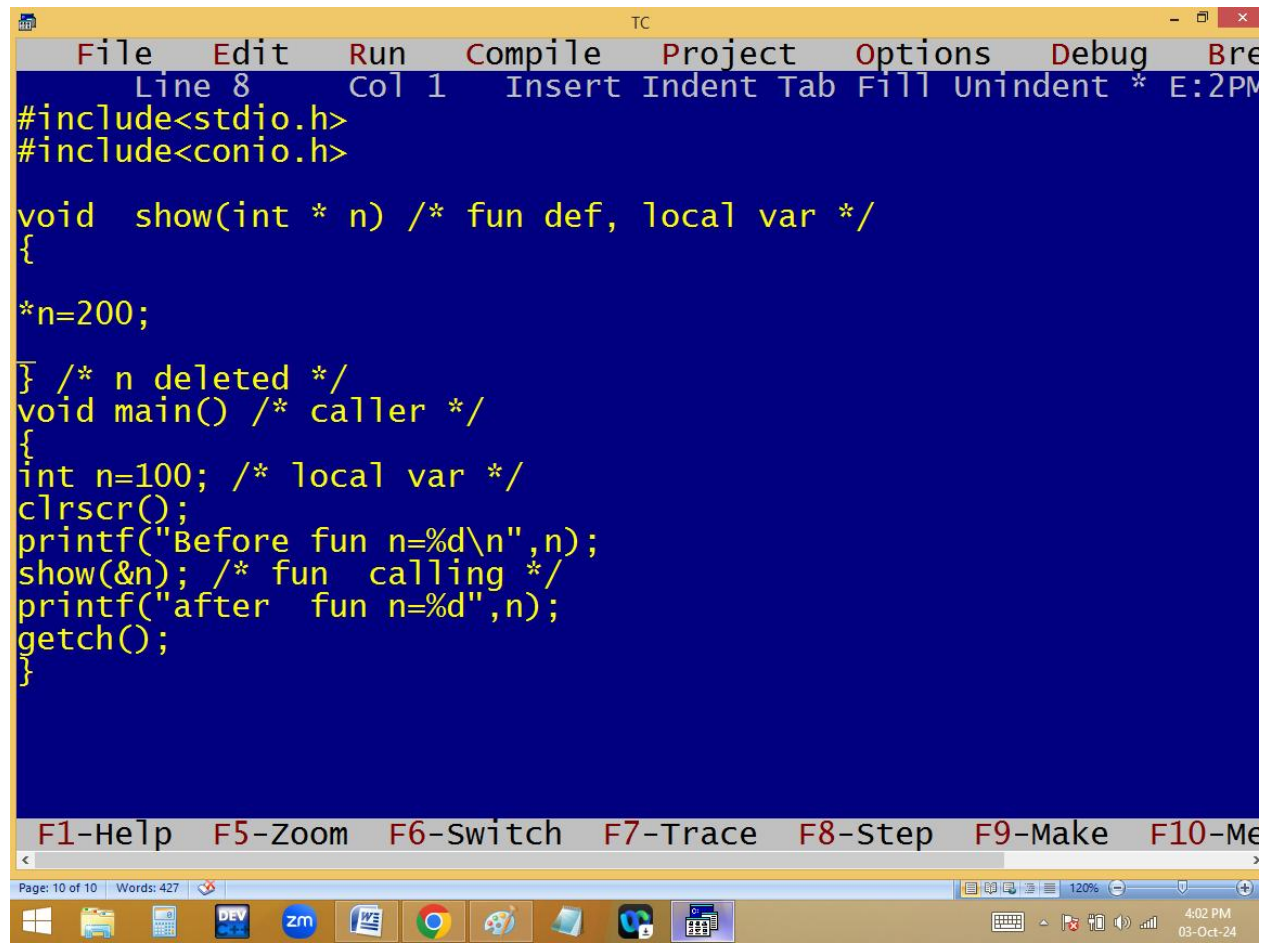
call / pass by address



\*x = 200

x value is 65500

\* means value at 65500 = 200

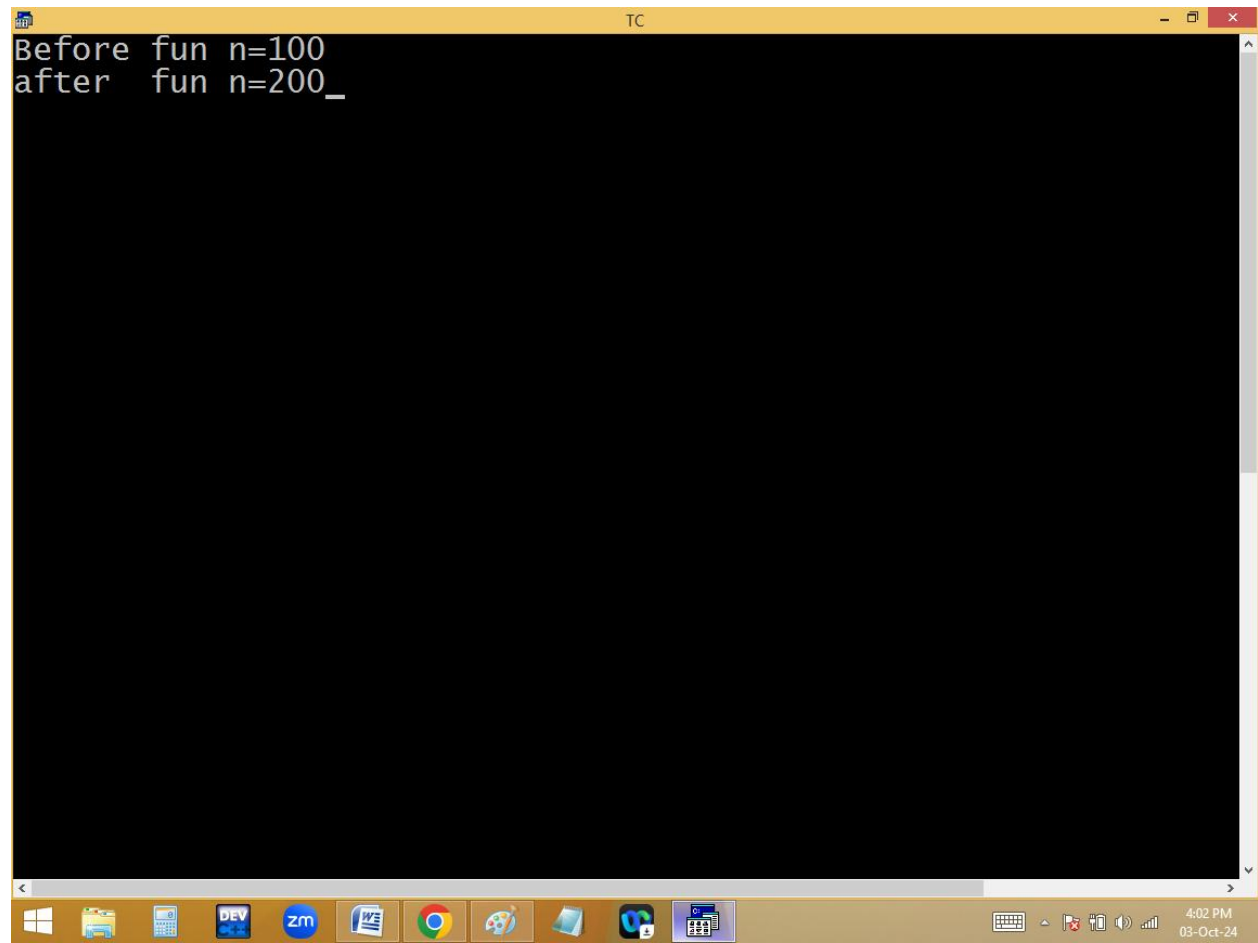


The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top indicates 'Line 8 Col 1' and 'Insert Indent Tab Fill Unindent \* E:2PM'. The code editor contains the following C program:

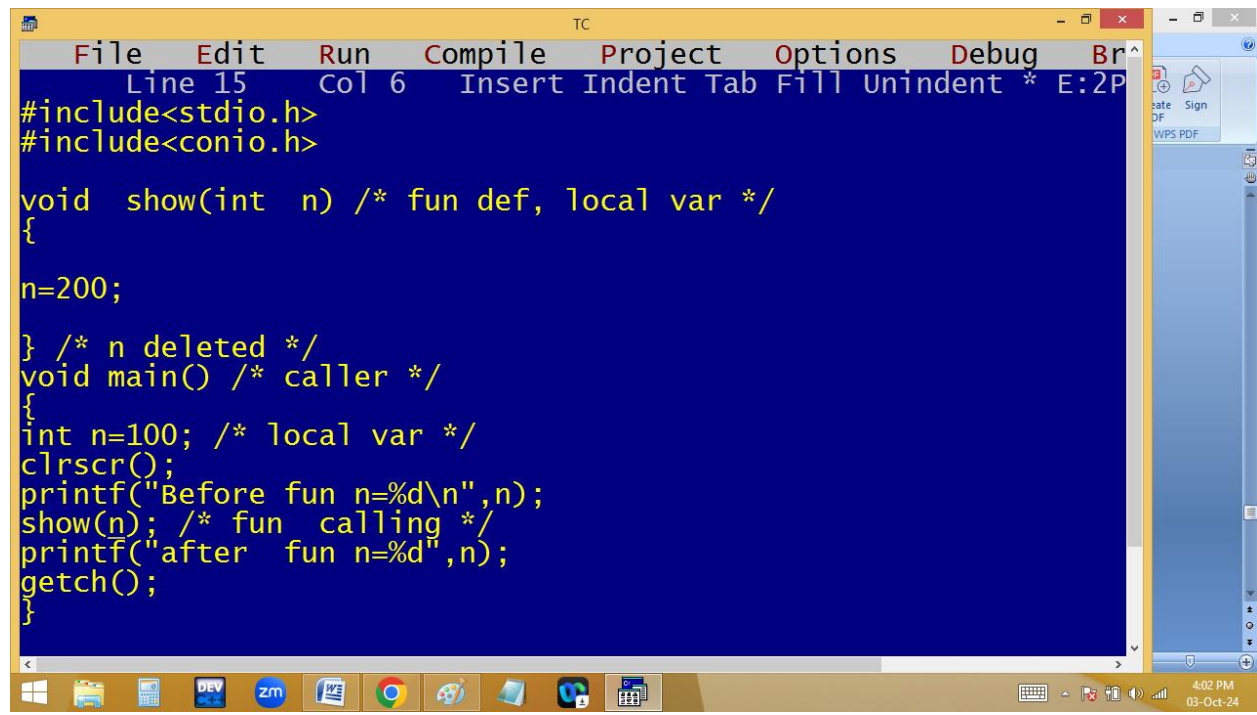
```
#include<stdio.h>
#include<conio.h>

void show(int * n) /* fun def, local var */
{
    *n=200;
} /* n deleted */
void main() /* caller */
{
    int n=100; /* local var */
    clrscr();
    printf("Before fun n=%d\n",n);
    show(&n); /* fun calling */
    printf("after fun n=%d",n);
    getch();
}
```

The bottom status bar shows keyboard shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Me. The taskbar at the bottom displays the Windows Start button, taskbar icons for File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and other applications. The system tray shows the time as 4:02 PM on 03-Oct-24.



```
Before fun n=100
after fun n=200_
```

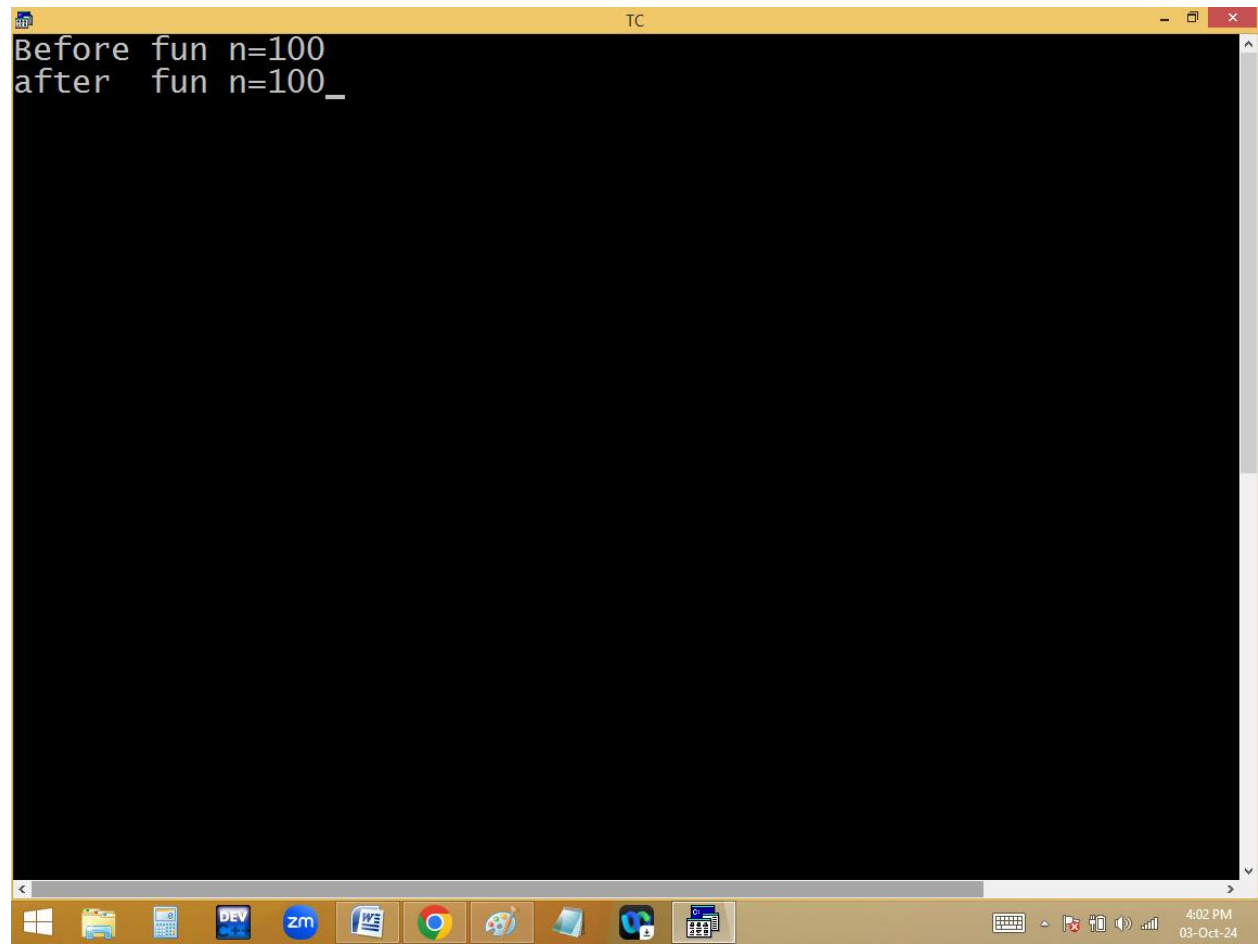


The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Breakpoint. The status bar at the top indicates 'Line 15', 'Col 6', and 'Insert' mode. The code editor has a dark blue background with yellow text. The code defines a function 'show' and a 'main' function. The 'show' function takes an integer 'n' and prints its value. In 'main', a local variable 'n' is set to 100, then 'show(n)' is called, and finally 'n' is printed again. The Windows taskbar at the bottom shows various application icons and the system clock indicating 4:02 PM on 03-Oct-24.

```
File Edit Run Compile Project Options Debug Br
Line 15 Col 6 Insert Indent Tab Fill Unindent * E:2P
#include<stdio.h>
#include<conio.h>

void show(int n) /* fun def, local var */
{
n=200;

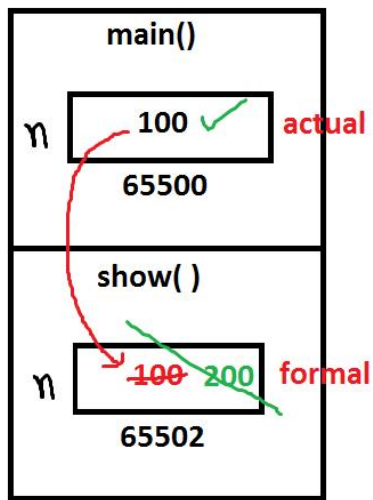
} /* n deleted */
void main() /* caller */
{
int n=100; /* local var */
clrscr();
printf("Before fun n=%d\n",n);
show(n); /* fun calling */
printf("after fun n=%d",n);
getch();
}
```



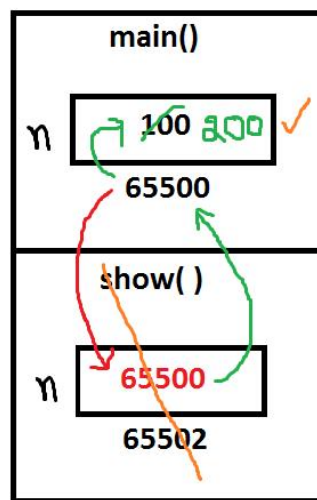
```
Before fun n=100
after fun n=100_
```

text/code area

call by value/ pass by value



call by addr/pass by addr



\*n = 200;  
↓  
\* 65500  
↓  
\* means value at that address  
\*65500==>value at 65500=200  
i.e. 100=200